

SolarGrid Demo Utility — Battery Energy Storage System Engineering Manual

SG-BESS-003

SOLARGRID AI — Synthetic Engineering Demonstration Corpus

Document ID	SG-BESS-003
Document Title	SolarGrid Demo Utility — Battery Energy Storage System Engineering Manual
Version	1.0
Revision Date	2026-01-15
Owner	SolarGrid Demo Utility — Engineering Department
Status	SYNTHETIC DEMO
Classification	DEMO / NON-REGULATORY
Purpose	Define the synthetic BESS architecture, operating limits, safety constraints, and evidence requirements.

Demo Note: This document contains synthetic engineering requirements created for the SOLARGRID AI demonstration. All numerical values, thresholds, equipment ratings, and procedures are illustrative and must be replaced or validated by the deploying organization before operational use.

Use restriction: This document is intended only for RAG testing, semantic retrieval testing, SafetyAgent demonstration, engineering evidence demonstration, PlannerAgent integration testing, and Tool 16 testing. It is not intended for real-world grid operation.

Document Control

Field	Value
Document ID	SG-BESS-003
Title	SolarGrid Demo Utility — Battery Energy Storage System Engineering Manual
Version	1.0
Revision Date	2026-01-15
Owner	SolarGrid Demo Utility — Engineering Department
Status	SYNTHETIC DEMO
Classification	DEMO / NON-REGULATORY

Scope: Define the synthetic BESS architecture, operating limits, safety constraints, and evidence requirements.

Revision History

Revision	Date	Description	Author
1.0	2026-01-15	Initial synthetic demonstration release	SolarGrid Demo Utility — Engineering Department

Table of Contents

- 1. 1. Purpose and Scope
- 2. 2. BESS Architecture
- 3. 3. Power and Energy Ratings
- 4. 4. State of Charge (SOC)
- 5. 5. Efficiency
- 6. 6. Temperature Limitations
- 7. 7. Degradation Considerations
- 8. 8. Emergency Shutdown and Safety Constraints
- 9. 9. Frequency Response
- 10. 10. Voltage Support
- 11. 11. Operating Reserve Contribution
- 12. 12. Availability Requirements
- 13. 13. Battery Safety and Evidence

1. Purpose and Scope

This engineering manual defines the synthetic BESS model used by the SolarGrid Demo Utility demonstration. It provides retrievable evidence for BESS power, energy, SOC, reserve, efficiency, temperature, safety, frequency response, voltage support, and availability constraints.

BESS values are synthetic demonstration parameters. They must be replaced or validated against the deploying organization's approved battery, inverter, protection, fire-safety, controls, and operating documentation before any real-world use.

2. BESS Architecture

The demonstration BESS is connected at the 132 kV network level through a transformer and bidirectional power-conversion system. The energy storage system consists of battery modules, a battery management system (BMS), power conversion system (PCS), supervisory controls, protection functions, and thermal monitoring.

BESS-001 — The demonstration BESS usable energy capacity is 120 MWh.

BESS-002 — The BESS is connected to the synthetic network at the N5 BESS 132 kV bus.

3. Power and Energy Ratings

Parameter	Synthetic Value	Engineering Interpretation
Usable energy capacity	120 MWh	Energy available within the approved operating window
Maximum continuous discharge	50 MW	Upper continuous active-power discharge limit
Maximum continuous charge	40 MW	Upper continuous active-power charge limit
Normal minimum SOC	20%	Lower normal operating boundary
Normal maximum SOC	90%	Upper normal operating boundary
Emergency reserve SOC	15%	Reserved energy fraction for emergency response

BESS-003 — The maximum continuous BESS discharge power shall not exceed 50 MW.

BESS-004 — The maximum continuous BESS charging power shall not exceed 40 MW.

BESS-005 — The BESS shall not be dispatched below the configured minimum SOC.

4. State of Charge (SOC)

BESS-006 — During normal grid operation, the BESS SOC shall remain between 20% and 90%.

BESS-007 — At 120 MWh usable capacity, 20% SOC corresponds to 24 MWh and 90% SOC corresponds to 108 MWh.

BESS-008 — At least 15% of usable BESS energy capacity shall be reserved for emergency response unless an approved operating exception is active.

Therefore, the synthetic emergency reserve energy is 18 MWh. Reserve SOC is not a substitute for the normal minimum SOC; both constraints must be checked.

5. Efficiency

Efficiency Parameter	Synthetic Value
Charge efficiency	94%
Discharge efficiency	94%

Efficiency Parameter	Synthetic Value
Round-trip efficiency	88% nominal planning assumption

BESS-009 — Energy accounting shall apply the configured charge and discharge efficiency assumptions rather than treating the BESS as lossless.

6. Temperature Limitations

BESS-010 — The BESS shall remain within its configured synthetic operating temperature band of 10°C to 40°C for normal operation.

BESS-011 — If monitored temperature exceeds 40°C, the BESS shall enter a restricted operating state pending thermal assessment.

BESS-012 — If a critical temperature or thermal-protection alarm is active, BESS dispatch shall be inhibited unless an approved emergency procedure explicitly authorizes a limited response.

7. Degradation Considerations

Battery degradation is represented as a planning consideration rather than a fixed life guarantee. Repeated high-power cycling, sustained high SOC, and operation near thermal boundaries may reduce available capacity over time.

BESS-013 — The planning model shall use the currently approved usable-energy capacity rather than assuming the original nameplate capacity remains available indefinitely.

8. Emergency Shutdown and Safety Constraints

BESS-014 — The BESS shall enter an emergency shutdown or inhibited state when a configured critical protection condition is active.

BESS-015 — A BESS emergency shutdown condition shall remove the affected BESS power capability from the available reserve calculation until the asset is explicitly returned to service.

9. Frequency Response

BESS-016 — The BESS may provide synthetic frequency-response reserve when SOC, power headroom, protection status, and availability permit.

BESS-017 — Frequency-response dispatch shall not violate the 50 MW discharge limit or the configured minimum SOC.

10. Voltage Support

BESS-018 — The BESS inverter may provide synthetic reactive-power voltage support when enabled and when the applicable network validation confirms sufficient inverter capability.

BESS-019 — Voltage-support operation shall not be represented as evidence that active-power, SOC, or thermal constraints are automatically satisfied.

11. Operating Reserve Contribution

BESS-020 — Eligible BESS reserve shall be limited by available discharge power, SOC above the applicable reserve floor, availability state, and response capability.

Example: if the BESS is at 80% SOC and delivering 30 MW discharge, the remaining continuous discharge headroom is 20 MW before considering any other constraints. The planner must still check reserve-energy requirements.

BESS-021 — A BESS reserve contribution shall not exceed 50 MW and shall not consume the protected emergency reserve energy unless an approved exception is active.

12. Availability Requirements

BESS-022 — A BESS availability state shall explicitly identify Available, Restricted, Unavailable, or Emergency Shutdown.

BESS-023 — A change in BESS availability greater than 10 MW of active-power capability shall trigger plan-impact assessment.

13. Battery Safety and Evidence

BESS-024 — Material BESS constraints used by a proposed plan shall have traceable engineering evidence in the approved evidence set.

BESS-025 — If required BESS evidence is missing or unverifiable, the validation result shall be UNKNOWN and shall require human review.

The BESS evidence model intentionally separates physical limits (power, energy, SOC, temperature) from operational decisions (dispatch, reserve allocation, and approval). This supports RAG retrieval and SafetyAgent reasoning.